

Same Data, Different Bins

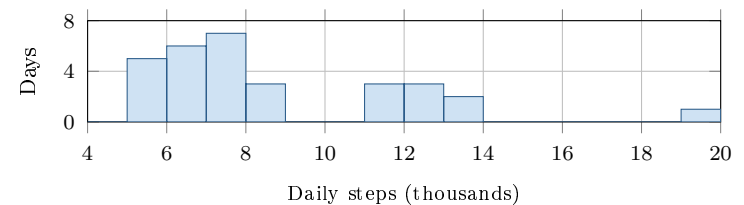
Unit 1 · Topic 1.5 · THE PROBLEM

Suppose a student recorded steps for 30 days. Histograms F and C show the same data with 1-thousand- and 5-thousand-step bins. A coach asks, “Does this student move differently on school days and weekends?”

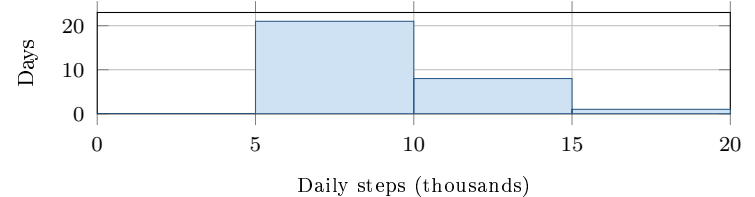
Priya: “Use C. F has too many short and empty bars; C gives the clearer summary.”

Leo: “Use F. C combines values that should not be treated as one group.”

F: 1-thousand-step bins



C: 5-thousand-step bins



FIRST TAKE — one sentence before you work the parts

Which histogram is more useful to the coach, F or C? Name one visible feature.

- DOK 1** (a) State each bin width and the number of days in F’s 19–20-thousand interval.
- DOK 2** (b) In two sentences, describe where F’s days are grouped and where its empty intervals are. Use thousands of steps.
- DOK 3** (c) Choose Priya’s or Leo’s display for a first look at the coach’s question. Cite one feature that the other display hides and explain how that changes the impression. What information would you still need to tell school days from weekends?

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.